

1. Artificial Intelligence?

Artificial Intelligence (AI) is one of the most important technologies of the modern world. It refers to the ability of machines and computer systems to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, making decisions, and adapting to new situations.

The concept of AI was introduced to create machines that can mimic human thinking and behavior. Instead of simply following fixed instructions, AI systems can analyze data, learn from experience, and improve their performance over time.

Today, AI is used in many aspects of everyday life. When we use voice assistants such as Siri, Alexa, or Google Assistant, AI helps them understand our commands and provide useful responses. Online shopping websites use AI to recommend products based on customer preferences. Navigation applications like Google Maps use AI to find the fastest routes and predict traffic conditions.

Features of Artificial Intelligence

Reasoning – AI can analyze information and make logical decisions.

Problem Solving – AI can identify solutions to complex problems.

Automation – AI can perform repetitive tasks without human intervention.



2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence has several branches and categories. Understanding the differences between AI, Machine Learning, Generative AI, and Agentic AI is important because each serves a different purpose.

1. Artificial Intelligence (AI)

Artificial Intelligence is the broad field that focuses on creating machines capable of performing tasks that typically require human intelligence. AI includes many technologies and techniques that allow computers to solve problems, recognize patterns, and make decisions.

Examples:

Voice assistants

Self-driving cars

2. Machine Learning (ML)

Machine Learning is a subset of AI. It enables computers to learn from data without being explicitly programmed for every situation. Instead of following fixed rules, machine learning models identify patterns in data and make predictions.

Examples:

Movie recommendations

Weather forecasting

3. Generative AI

Generative AI is a specialized area of AI that focuses on creating new content. It can generate text, images, videos, music, and computer code. Generative AI learns patterns from large datasets and produces original outputs based on user prompts.

Examples:

ChatGPT generating text

AI image generators creating artwork

AI tools generating programming code

4. Agentic AI

Agentic AI refers to intelligent systems that can independently plan, make decisions, and execute tasks to achieve goals. These systems can perform multiple actions with minimal human supervision.

Examples:

AI assistants managing schedules

Autonomous robots

AI agents performing research tasks

Autonomous Agent

3. Explain Any Three Open-Source AI Models Available on Hugging Face

Hugging Face is one of the world's largest platforms for sharing and accessing AI models. It provides thousands of open-source models for research, learning, and development.

1. Llama 3

Llama 3 is an advanced open-source large language model developed by Meta. It is designed to understand and generate human-like text. Llama 3 can answer questions, write essays, generate code, summarize information, and assist with learning tasks.

Features

High-quality text generation

Applications

Chatbots

Content creation

Software development

2. Mistral 7B

Mistral 7B is a powerful and efficient language model developed by Mistral AI. Despite having fewer parameters than some larger models, it delivers excellent performance while requiring fewer computing resources.

Features

Lightweight architecture

Fast response generation

3. FLAN-T5

FLAN-T5 is an instruction-tuned language model developed by Google. It is trained to follow user instructions effectively and perform a variety of natural language processing tasks.

Features

Strong instruction-following ability

Multitask learning capability

Supports language understanding

Good text summarization performance

Applications

Question answering

Language translation

Text summarization

Educational tools

4. What is the Difference Between Traditional Programming and AI-Assisted Development?

Software development has evolved significantly with the introduction of Artificial Intelligence. Traditional programming and AI-assisted development represent two different approaches to creating software solutions.

Traditional Programming

In traditional programming, developers manually write all instructions and logic required for a program. The computer follows these instructions exactly as written.

Process

Define the problem.

Write algorithms.

Develop code manually.

Test and debug.

Deploy the application.

Characteristics

Rule-based approach.

Every condition must be programmed.

Requires significant coding effort.

Less flexibility.

Example

If a programmer wants to create a calculator, they must manually write code for addition, subtraction, multiplication, and division operations.

AI-Assisted Development

AI-assisted development uses AI tools to support programmers during software creation. AI can suggest code, detect bugs, explain concepts, generate documentation, and improve productivity.

Features

Automatic code suggestions.

Error detection.

Code completion.

Documentation generation.

Examples of AI Tools

GitHub Copilot

ChatGPT

Conclusion

Traditional programming remains essential because developers must understand programming concepts and make final decisions. However, AI-assisted development enhances efficiency by providing intelligent support throughout the development process.

5. Explain AI Hallucinations with One Example

AI hallucination is a phenomenon where an AI system generates information that appears correct and convincing but is actually false, inaccurate, or completely fabricated. Hallucinations occur because AI models generate responses based on patterns learned from training data rather than genuine understanding.

An AI system may confidently provide incorrect information when:

It lacks sufficient knowledge.

The question is ambiguous.

Training data is incomplete.

The model predicts incorrect patterns.

Example of AI Hallucination

Suppose a person asks:

"Who was the first human to land on Mars?"

Since no human has landed on Mars yet, the correct answer should be that no person has achieved this milestone.

However, an AI system experiencing hallucination might respond:

"The first person to land on Mars was John Smith in 2030."

This answer sounds realistic but is completely false because no human mission to Mars has taken place.

Types of AI Hallucinations

1. Factual Hallucination

Providing incorrect facts or information.

Example: Giving the wrong date for a historical event.

2. Citation Hallucination

Creating fake references, books, or research papers.

Example: Inventing a scientific article that does not exist.

3. Logical Hallucination

Producing reasoning that appears logical but contains errors.

Example: Drawing incorrect conclusions from correct data.

Conclusion

AI hallucinations are incorrect or fabricated responses generated by AI systems. Although AI can provide useful information, users must carefully evaluate and verify AI-generated outputs before relying on them for important decisions.